Yupei Duan

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Research interest: Designing AI-driven immersive systems to enhance STEM learning.

Education

Degree	Institution and field	Dates
Ph.D.	University of Missouri, Columbia, Missouri, USA — *Information Science and Learning Technologies*	2023–July 2028 (expected)
M.Sc.	University of Missouri, Columbia, Missouri, USA — *Information Science and Learning Technologies*	2019–2021
B.Sc.	Capital Normal University, Beijing, China — *Biological Sciences*	2002–2006

Professional Experience

Future Leadership Academy, Hainan, China
Director, Curriculum Development Center; Academic Dean; Science Teacher
2017–2023

Beijing No. 4 High School, Beijing, China
Director of Middle School, International Division; Director of Science Education, Junior High Division; Science Teacher; Biology Teacher
2006–2017

Research Experience

University of Missouri, Columbia, Missouri, USA

- Research Assistant, Technology-Enhanced Learning Lab, 2023–Present
- Research Assistant, Information Experience Lab, 2023–Present
- Research Assistant, Build It Green Project, 2023–2025

Peer-Reviewed Journal Articles

Duan, Y., Xu, X., Wang, F., Franken, O., Gu, Y., Li, S., & Almasri, M. (under review). From scaffolds to learner-initiated AI: Redistributing responsibility in photolithography training. *Educational Technology & Society*.

Duan, Y., Xu, X., Wang, F., Stafford, A., Murrell, S., Li, S., & Aldrich, E. (under review). Meeting the clouds in your hand: Using augmented reality to facilitate learning in a university meteorology class. *Journal of Computer Assisted Learning*.

Duan, Y., Xu, X., He, H., Gu, Y., Li, S., Bueno-Vesga, J., & Fu, D. (under review). When virtual patients speak: From realism to learner agency in AI-driven VR standardized patient encounter training. *Computers & Graphics*.

He, H., Xu, X., Li, S., Bueno-Vesga, J. A., **Duan, Y.**, & Gu, Y. (2026). Training nursing skills in a generative artificial intelligence-enhanced virtual reality patient encounter simulation: A qualitative study from a student perspective. *International Journal of Educational Technology in Higher Education, 23*, Article 12. https://doi.org/10.1186/s41239-026-00587-9

Duan, Y., Xu, X., He, H., Li, S., & Gu, Y. (2025). Blending mixed reality and generative AI to teach geography: An MR+GenAI learning environment. *Journal of Interactive Learning Research, 36*(1), 83–100. https://www.learntechlib.org/primary/p/225218/

Bueno-Vesga, J. A., Xu, X., Gu, Y., He, H., Li, S., & **Duan, Y.** (2026). The role of cognitive load and engagement in students' adoption of an AI-supported VR training system. *IEEE Transactions on Visualization and Computer Graphics, 32*(5), 4617–4625. https://doi.org/10.1109/TVCG.2026.3680684

Duan, Y., & Chen, Y. (2015). Construct models of the human circulatory system to improve teaching and learning. *Bulletin of Biology, 50*(10), 33–35, 63. CNKI:SUN:SWXT.0.2015-10-013. [In Chinese]

Duan, Y. (2015). The golden key to the world of life: Content related to inquiry skills in the BNU edition biology textbook. *Journal of Chinese Foundation Curriculum*, (3), 62–64. [In Chinese]

Duan, Y., Wang, Y., Li, J., & Gong, Y. (2013). The rising stage of science: Developing the school-based elective course "Science Around Us" through collaboration among teachers from different disciplines. *Beijing Education*, (4), 42–44. [In Chinese]

Chen, Y., Lei, M., & **Duan, Y.** (2010). Using school-based laboratory reports to improve the efficiency of middle school experimental instruction. *Bulletin of Biology, 45*(4), 42–45. [In Chinese]

Duan, Y. (2009). Using blogs to improve middle school biology teaching and learning. *Bulletin of Biology, 44*(4), 42–44. [In Chinese]

Conference Proceedings

Li, S., Xu, X., Wang, F., **Duan, Y.**, & Mahmoud, A. (2026). Guiding the redesign of VR-based photolithography training from domain experts' perspectives: A think-aloud study. *Immersive Learning Research—Academic, 1*(1). Publisher record

Duan, Y., Xu, X., He, H., Gu, Y., Li, S., & Bueno-Vesga, J. (2026). Supplementing patient encounter training for preservice nurses: Toward an AI×VR approach. *2026 IEEE International Conference on Artificial Intelligence and Extended and Virtual Reality (AIxVR)*, 98–107. **Best Paper Nominee.** https://doi.org/10.1109/AIxVR67263.2026.00020

Duan, Y., Xu, X., Wang, F., Shyu, C.-R., Islam, S. K., Eliza, S., Flink, J., Gu, Y., He, H., Li, S., Murrell, S., Nalmas, A., & Almasri, M. (2025). Enhancing student self-efficacy and interest in microelectronics through immersive virtual reality in an informal learning environment. *2025 IEEE International Conference on Advanced Learning Technologies (ICALT)*, 308–312. https://doi.org/10.1109/ICALT64023.2025.00096

Xue, L., Glaser, N., & **Duan, Y.** (2025). Doctoral colloquium—Narrative and soundscapes: Crafting a video game at the intersection of neurodiversity, gender, and queerness—*The Things Left Behind*. *Immersive Learning Research—Academic, 1*(1), 103–109. https://doi.org/10.56198/0efasn65

Glaser, N., Calyam, P., **Duan, Y.**, Li, S., Nuguri, S. S., Ousley, C., Kambhampati, A., Parishani, Z., Joshi, A. C., & Yang, M. (2025). Preliminary analysis of empathy-driven design and inclusive cybersecurity education: The initial phase of the uSucceed project's virtual reality curriculum for neurodiverse adults in STEM. In J. M. Krüger et al. (Eds.), *Immersive Learning Research Network* (Communications in Computer and Information Science, Vol. 2271). Springer. https://doi.org/10.1007/978-3-031-80475-5_19

Books and Contributed Works

Han, Q., et al. (2016). *Hundred Thousand Whys: Life* (6th ed.). Juvenile & Children's Publishing House. ISBN 978-7-5324-9801-7. **Contribution: Listed co-author.**

Conference Presentations

Duan, Y., & Oprean, D. (2026, November). *Asking good questions in AI-supported problem solving: A scoping review of AI literacy and information behavior in education* [Accepted paper presentation]. Association for Library and Information Science Education (ALISE) 2026 Annual Conference, Buffalo, NY, USA.

Xue, L., Fu, D., **Duan, Y.**, & Moore, J. (2026, November). *Time2Reflect: A multi-agent AI system for scaffolded post-class reflection in STEM graduate education* [Accepted paper presentation]. Association for Library and Information Science Education (ALISE) 2026 Annual Conference, Buffalo, NY, USA.

Duan, Y., & Oprean, D. (2026, November). *Bridging design thinking and artificial intelligence literacy through phase-level mapping: A scoping review* [Presented] paper presentation. Association for Educational Communications and Technology (AECT) International Convention, Chicago, IL, USA.

AERA 2027 Annual Meeting · Accepted paper presentation
From Virtual Practice Toward Cleanroom Readiness: Productive Effort and Usability Friction in VR Training
Toronto, ON · April 14–18, 2027 · Paper Session: From Simulation to Practice: Skill Transfer in the Professions

Duan, Y., Xu, X., He, H., Bueno-Vesga, J., Gu, Y., & Li, S. (2026, November). *From desktop to immersive VR in nursing education: Design-based research on system adaptation* [Presented] paper presentation. Association for Educational Communications and Technology (AECT) International Convention, Chicago, IL, USA.

Duan, Y., Xu, X., Bueno-Vesga, J., Li, S., Cox, M., Gu, Y., Fowler, S.-Y., & Claunch, H. (2026, April). *"Let's practice together": Exploring collaborative virtual reality for nursing students' IVPB medication administration training* [Paper presentation]. American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA, USA.

Rende Mendoza, K., **Duan, Y.**, Churaman, T., Ogundairo, D., Glaser, N., Jensen, L. J., Olsen, A. A., & Schmidt, M. (2026, April). *Reimagining CT and STEM learning with neurodivergent youth through rightful presence and pedagogies of play* [Paper presentation]. National Association for Research in Science Teaching (NARST) Annual International Conference, Seattle, WA, USA.

Duan, Y., Xu, X., Wang, F., Shyu, C.-R., Islam, S. K., Eliza, S., Flink, J., Gu, Y., He, H., Li, S., Murrell, S., Nalmas, A., & Almasri, M. (2025, October). *Nurturing tomorrow's engineers: Strengthening self-efficacy and interest in microelectronics through virtual reality* [Paper presentation]. Association for Educational Communications and Technology (AECT) International Convention, Las Vegas, NV, USA.

Duan, Y., Xu, X., Gu, Y., Li, S., Fowler, S.-Y., Claunch, H., Snyder, J., He, H., & Bueno-Vesga, J. (2025, April). *The usability of an artificial intelligence-enhanced virtual standardized patient (AI-VSP) in nursing* [Roundtable presentation]. American Educational Research Association (AERA) Annual Meeting, Denver, CO, USA.

Snyder, R., Webb, C., **Duan, Y.**, Zangori, L., & Cole, L. (2025, April). *The integration of place in rural science education: Resources of place realized by middle school teachers through the implementation of a place-based engineering design unit* [Paper presentation]. American Educational Research Association (AERA) Annual Meeting, Denver, CO, USA.

Justice, J., **Duan, Y.**, Webb, C., & Zangori, L. (2024, November). *Build It Green: A new place-based energy unit for science educators* [Paper presentation]. Science Teachers of Missouri 2024 Conference, Columbia, MO, USA.

Duan, Y., Xu, X., Li, S., Xue, L., He, H., & Gu, Y. (2024, October). *Movement matters: How a virtual reality learning system facilitates U.S. geography learning for international students* [Paper presentation]. Association for Educational Communications and Technology (AECT) International Convention, Kansas City, MO, USA.

Glaser, N., **Duan, Y.**, Calyam, P., Li, S., Nuguri, S. S., Kambhampati, A., Parishani, Z., & Singh, K. P. (2024, October). *Co-design of a virtual reality cybersecurity training platform for neurodiverse adults* [Paper presentation]. Association for Educational Communications and Technology (AECT) International Convention, Kansas City, MO, USA.

Glaser, N., Calyam, P., **Duan, Y.**, Li, S., Nuguri, S. S., Ousley, C., Kambhampati, A., Parishani, Z., Joshi, A. C., & Yang, M. (2024). *Preliminary analysis of empathy-driven design and inclusive cybersecurity education: The initial phase of the uSucceed project's virtual reality curriculum for neurodiverse adults in STEM* [Paper presentation]. 10th International Conference of the Immersive Learning Research Network (iLRN 2024).

Duan, Y. (2018, April 12–14). *Another song of ice and fire: A teacher's findings, reflections, and actions related to STEAM education in China* [Workshop]. International Technology and Engineering Educators Association 80th Annual Conference, Atlanta, GA, USA.

Duan, Y. (2017, March 25–26). *Introduction to project-based learning at Beijing No. 4 High School International Division* [Professional development workshop]. OESIS–CERNET Beijing, Beijing, China.

Science Communication and Popular Science Publications

Jenson-Elliott, C. (2022). *Weeds Find a Way*. Beijing United Publishing. ISBN 978-7-5596-6228-6. **Chinese translator: Yupei Duan.**

Gonick, L., & Wessner, D. (2019). *The Cartoon Guide to Biology*. China Science and Technology Press/Popular Science Press. ISBN 978-7-1100-9979-7. **Chinese translator: Yupei Duan.**

Kelly, R. (2017). *Ultimate Guide: Ocean*. Miles Kelly. ISBN 978-7-5315-8178-9. **Chinese translator: Yupei Duan.**

Duan, Y. (2015–2021). "Play Science Games at Home" [Popular science columnist]. *Knowledge Is Power*. Popular-science archive [In Chinese]

Program Designer and Host. *Four-Dimensional Science Masters* (四维科学人), China Central Television. Selected science programs [In Chinese]

Program Designer and Host. *Science Show* (科学达人秀), China Central Television. Selected science programs [In Chinese]

Program Designer. *Miracle Numbers* (心里有数), China Central Television. Selected science programs [In Chinese]

Journal and Conference Reviewing

- Reviewer, American Educational Research Association (AERA) 2027 Annual Meeting, SIG–Advanced Technologies for Learning — 2 proposals, August 2026
- Reviewer, *BMC Nursing* — 1 manuscript, July 2026
- Reviewer, Association for Educational Communications and Technology (AECT) International Convention — 6 proposals, May 2026
- Reviewer, Association for Educational Communications and Technology (AECT) International Convention — 2 proposals, May 2025
- Reviewer, 25th IEEE International Conference on Advanced Learning Technologies — 31 submissions, February 2025
- Reviewer, Association for Educational Communications and Technology (AECT) International Convention — 7 proposals, May 2024

Teaching

University of Missouri

Term	Course	Role
Fall 2026	IS_LT 2310: Technologies for Game-Inspired Teaching and Learning	Instructor
Fall 2026	IS_LT 2300: Inquiry into Transforming Learning with Technology	Instructor
2026	IS_LT 7361: Introduction to Digital Media	Teaching Assistant
Fall 2026	IS_LT 7888: Introduction to VR-enabled Instructional Design	Teaching Assistant

Future Leadership Academy

Dates	Course	Role
2019–2023	Instructional Design and Technology	Instructor
2019–2023	Digital Learning	Instructor
2017–2023	Biology	Instructor

Beijing No. 4 High School

Dates	Course	Role
2008–2017	Science Around Us	Instructor
2006–2017	Biology	Instructor

Honors and Awards

Award	Academic year
London Recognition Scholarship Fund, University of Missouri	2023–2027
Jonassen Endowment in Information Science, SISLT, University of Missouri	2023–2024

Professional Affiliations

Organization	Dates
Association for Library and Information Science Education (ALISE)	2026–Present
National Association for Research in Science Teaching (NARST)	2026–Present
Institute of Electrical and Electronics Engineers (IEEE)	2025–Present
American Educational Research Association (AERA)	2024–Present
Association for Educational Communications and Technology (AECT)	2023–Present

Selected Professional Development

Program	Date
CITI-Sci Program, Mizzou TechHub; sponsored by the U.S. National Science Foundation · **CITI-Sci badge**	**November 2026**
Engagement, Equity, and Inclusion Workshop, Teaching for Learning Center, University of Missouri	August 2023
Certificate in School Management and Leadership, Harvard Graduate School of Education and Harvard Business School Online	August 2021
IB MYP Sciences: Delivering the MYP Curriculum, Category 2, International Baccalaureate Organization	November 2019
Diversity and Inclusion Certificate, Cornell University ILR School via eCornell	May 2019
IB PYP Assessment, International Baccalaureate Organization	November 2018
IB MYP Category 1 Workshop, International Baccalaureate Organization	July 2018
IB PYP Administrator Certification, International Baccalaureate Organization	June 2018
Instructional Design Training, National Normal Universities Academic Forum on Biology Teaching	November 2016
Maker Education Mentor Certificate, Tsinghua University	June 2016

Technical Skills

- **Data analysis and programming:** R, JavaScript
- **Immersive and interactive development:** Unity, OpenSimulator, Twine, Construct 3, HTML, CSS
- **Research, design, and media tools:** Adobe Creative Cloud, Microsoft Office, Adobe Captivate, Lucidchart
- **Learning platforms:** Canvas, Moodle, VoiceThread